

Perceptions of Agility and Collaboration in Software Development Practice

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Abstract

Agility is a facet of software development attracting increasing interest. The purpose of this paper is to investigate the value of agility in practice. Data collected from senior development/project managers in 62 organizations is used to investigate perceptions related to agile development. Specifically, the perceptions tested relate to (a) belief in agile values and principles, and (b) value of agile principles within current development and organizational practice. These perceptions are then examined in the context of current practice in order to test perceptions against behavior and understand the valued aspects of agile practice that are implicit in software development today. The findings suggest that agile values and principles related to communication and collaboration, and team involvement are perceived as important and are being practiced widely. On the other hand, principles related to reflection and frequent delivery of software are rated high to medium in importance but its full adoption in practice is limited. The principle relating to managing changing requirements is perceived as significant but agile practices relating to this principle are not being implemented in practice.

management practices for software engineering. Early efforts resulted in structured methods, which have evolved and expanded over time to embrace consortia/company-led initiatives such as the Unified Modeling Language and the Unified Process alongside formal process frameworks such as the International Standards Organization's 9000 series, the Capability Maturity Model and SPICE.

More recently, the philosophy behind engineering initiatives has been questioned by the agile movement, which seeks to emphasize the human and craft aspects of software development over and above the engineering aspects [5-7]. Agile practice is strongly collaborative in its outlook, favoring individuals and interactions over processes and tools, working software over comprehensive documentation, customer collaboration over contract negotiation, and responding to change over following a plan. Early experience reports on the use of agile practice suggest some success in dealing with the problems of the software crisis [8], and suggest that engineering and agile practice are not mutually exclusive [9]. With this duality in mind, this article surveys the current practice in software engineering alongside perceptions of senior development managers in relation to agile practice in order to understand the principles of agility that may be practiced implicitly.

1. Introduction

Organizations undertaking software development are often reminded that successful practice depends on a number of non-technical issues that are managerial, cultural and organizational in nature [1-3]. These issues cover aspects from appropriate corporate structure, through software process development and standardization to effective collaborative practice. Since the recognition of the 'software crisis' [4] in the late 1960s, significant effort has been put into addressing problems related to the cost, time and quality of software development via the application of systematic processes and

2. Background

In early 2001 a group of seventeen leading methodologists, known as the Agile Alliance, convened with the intention of formulating a set of values and principles for good software development practice. These values and principles place people, their skills and capabilities, and personal relationships at the heart of the software development process. In contrast with most part of the past software development literature, agile development shifts its focus away from methods and techniques as the main

thrust to delivering quality software. Agile software development (ASD) focuses on creating and efficiently responding to change in order to profit in a turbulent business environment [7, 10]. Increasingly, the approach has been adopted by the software development community as a viable solution to the problems of the ‘software crisis’ which are still a reality today (i.e., problems of budget and schedule overruns, low quality and unreliable systems, and non-conformance to users’ needs).

ASD places strong emphasis on collaboration and interaction through four core values. The values agreed by the Agile Alliance are: (1) *individuals and interactions* over processes and tools; (2) *working software* over comprehensive documentation; (3) *customer collaboration* over contract negotiation; and (4) *responding to change* over following a plan [5]. The methodologists of the Agile Alliance also agreed on twelve principles (see Table 2) that support the values and decided to leave the practical application of the principles to individuals.

Current agile software development research is driven by the two core ideas elicited from the agile values and principles by Highsmith and Cockburn [11]: (a) different projects need different processes or methodologies and (b) focusing on skills, communication, and community allows projects to be more effective and more agile than focusing on processes. ASD aims to reduce the cost of change throughout the project, hence reducing the cost of rework.

3. Details of the Survey

To elicit an understanding of current practice and perceptions related to agility, an online questionnaire was designed with the following objectives in mind:

- To review current software development practice.
- To assess perceptions related to agile principles and values.
- To examine the implicit/explicit integration of agile principles with mainstream development and organizational practice.

Based on these research objectives, a web-based survey technique was considered the most suitable research strategy for this study. A key feature of surveys is to “obtain snapshots of practice, situations or views at a particular point in time (via questionnaires or interviews) from which inferences are made (using quantitative analytical techniques)

regarding the relationship that exist in the past, present and future” [12].

Surveys have been widely used by researchers in software development research for the purpose of capturing current trends in practice [13-18]. Surveys have the potential of capturing large amounts of data from a wide sample in a short space of time. The advantage of having a wide sample can lead to greater validity as the sample can be more representative. Results thus can be generalized to a larger population. As Marshall and Rossman [19] aptly state: “Survey research is the appropriate mode of inquiry for making inferences about a large group of people from data drawn on a relatively small number of individuals from that group” [19]

E-mail was used to contact senior development managers at 970 organizations, requesting that they participate in the research. The selection of participants was based on the contact list available and research noting that senior management influence productivity [20]. The survey itself comprised of 32 questions that were organized into three sections addressing (a) background/demographics, (b) current software development practice and (c) perceptions related to agility (the questionnaire is available on request).

The background section of the questionnaire concerned questions related to respondents’ experience in the area of software development, their knowledge, their organization and their current project. These questions were aimed at validating the response sample and carrying out cross-analysis between groups.

The objective of the second section (Current Software Development Practice) was to understand the state-of-practice in relation to the use of software development methods and the current practice of agile principles in organizations. The section contained questions on the adoption of software development methods/approaches and the way that they are used in practice. Specific questions were asked on the current implementation of each agile principle through current practices (i.e., are agile practices implemented implicitly or explicitly?). These questions were created to correlate the gap between the respondents’ current implementation of current practice (i.e., what they do) to the respondents’ perception of agile principles (i.e., what they think).

The last section of the questionnaire contained questions specifically related to agile values and principles. The aim of such questioning was to (a) ascertain perceptions related to the value and importance of agility and (b) relate that information back to the context of current practice in order to understand the match between principle and practice.

In summary, respondents were presented with background information related to agile values and principles and asked to rank the principles in terms of importance from an organizational perspective and then rate its importance from a personal perspective.

Online presentation of the questionnaire ensured that the section on current practice had to be completed before the section on perceptions of agility could be viewed and completed.

A pilot study was carried out on 8 industrialists. The purpose of the pilot was to test (a) the understandability and clarity of questions, (b) the time it took to complete the questions and (c) to test the correct input of data collected by the web-base. Their comments were used to refine the questionnaire. The questionnaire was shortened and some questions were reworded for the purpose of clarity.

Table 1 summarizes the demographics of the 62 organizations that responded to the online survey (equating to a response rate of 6%). Overall, the respondents had on average more than 10 years development experience, had variation in their development experience (having worked in more than 4 software development roles), and worked in organizations that had more than 500 employees. The majority of responses received were from IT Directors, Senior Architects and IT Managers. In addition, responses from the larger organizations covered in-house development departments of a variety of sizes from fewer than 10 employees to more than 500 employees.

Table 1. Demographics of respondents

Business Category	% of Respondents	Average Development Experience	Average Number of Development Jobs Experienced	Current Job Title of Majority of Respondents	Average Organisation Size (employees)
Consultancy/ Software house	35%	More than 10 years	Greater than 4	Senior Architect	Greater than 500
Construction/ Manufacturing/ Distribution	12%	More than 10 years	Greater than 4	IT Manager	250 – 500
Finance/ Insurance/ Real Estate	12%	More than 10 years	Greater than 4	IT Director	Greater than 500
Government/ Public Sector/ Education	12%	More than 10 years	Greater than 4	IT Director	50 – 249
Media Services/ Communication	14%	6 – 10 years	Greater than 4	Project Manager	0 – 9
Whole Sale/ Retail trade	7%	More than 10 years	Greater than 4	Head of IT	Greater than 500
Charity	4%	6 – 10 years	Greater than 4	Head of IT	Greater than 500
Legal	4%	More than 10 years	Greater than 4	IT Director	Greater than 500
Overall Majority		More than 10 year (60%)	Greater than 4 (90%)	Senior Architect (28%)	Greater than 500 (64%)

4. General View of Software Development

The objective of understanding the state-of-practice was providing a benchmark in order to assess the gap with the concepts of agile practice. The use of software development process provided the framework for questioning and a number of questions were asked to provide correlation points for agile practice (discussed later). The major points of interest are that:

- Most organizations employ a software development process but implement it in a flexible/configured manner.
- Development processes are supported by the strong use of iterative lifecycles, but change is implemented in a structured manner that does not necessarily fit.

In detailed terms, more than half the sample claim to use a formal software development process within their organization, either developed in-house (56%) or commercially branded (18%). Of the commercially branded approaches, there was a fairly even utilization of the Rational Unified Process (RUP), Structured Systems Analysis and Design Methodology (SSADM) and the Dynamic Systems Development Methodology (DSDM). Interestingly, however, organizations using formal software development processes (a) tend to tailor/configure them strongly (82%), lending weight to the adage that ‘people don’t adopt a methodology, they adapt it’ (attributed to Tom DeMarco) and (b) use different processes for different types of project (79%).

The levels of effort put into tailoring/configuring process varied: 16% stated they spent 1% percent of overall project time on tailoring the software development process; 67% of respondents stated the time to tailor processes ranged from 1% to 5% of overall project time; 19% stated they spent 10% – 20% of the project time and 2% stated that they use 40% of the project time. The type of people involved in process tailoring were senior management, project managers, architects and the development team – where there appeared to be a decision maker, the development team was always involved in the decision. Lastly, ‘project characteristics’ were the predominant criteria on which tailoring decisions were made (44%).

Of the respondents that do follow some form of software development process, the majority stated that their processes are based on an iterative and/or incremental software development lifecycle (68%), the remainder being based on the more sequential waterfall model. Nearly all respondents stated that

they cater for changing requirements, even late in the project (98%). Of that proportion, only some follow the dynamic prioritization path proposed by iterative/incremental development (39%) - the majority of respondents continue to use traditional change control request (61%).

5. Perceptions of Agile Principles

The objectives of questioning around agile principles were (a) to ascertain perceptions related to the value and importance of agility and (b) to relate that information back to the context of current practice in order to understand the match between principle and practice. Summary results are shown in Table 2; these findings were then linked back to earlier questions to match perceptions with practice. The major points of interest are that:

- The most widely valued agile principles relate to the frequent delivery of working software, daily interaction between business people and developers and the importance of face-to-face communication.
- The highest levels of equivocation exist around principles related to changing requirements late in the development cycle and working software being the primary measure of progress.
- The lowest areas of organizational importance were that the needs of the development team must be prioritized in line with cost, quality and time and that architecture and design should emerge and not be imposed.

In detailed terms, the results showed Principles 1, 5 and 7 (see Table 2) as unequivocal in their perceived importance (60, 84 and 60% respectively). When examining the frequent delivery of working software the results from current practice showed that the majority of respondents delivered on three monthly cycles (31%). Significantly, however, a number of respondents delivered on monthly cycles (24%) and two-week cycles (24%) – smaller percentages delivered on weekly cycles (5%) or produced only a final release (17%). In communication terms, the frequency of communication between Business and IT ranged from approximately once a month (13%), through once or twice a week (56%), to communication on a daily basis (29%). The primary means of communication within software development practice is face-to-face within development team (79%) - few respondents use Email (17%) or telephone (4%) as a primary communication tool for example.

While the delivery of working software was seen as important, the results related to software being the primary measure of progress were much more ambiguous. Here, findings from the data collected in practice showed a strong reliance on traditional project metrics. While a number of project respondents do use working software as the primary measure of progress (26%), a significant number use other project milestones and deliverables (54%) or completion of process/lifecycle stages (20%). Similarly, while the majority of respondents state that they cater for change late in the development cycle, change does not appear to be a particularly welcome factor. Interestingly, however, there is a strong correlation between the respondents who embrace change (40%) and their use of dynamic change control techniques (38%).

Low areas of organizational importance indicate that a significant number of respondents (47%) value human facets of development over and above typical facets of cost, quality and time. This view is borne out to a given degree when the data is correlated with the view that projects should be built around motivated individuals (53%) and practice indicating that (a) the development team is involved with decision making (64% on average) and (b) individuals are empowered to make decisions on work directly related to them (89%). In contrast, however, the results note that, while the development team is involved in tailoring the software development process (64%), team skills are the least used tailoring criteria (22%).

Table 2. Agile perception results

	Principle	Importance for Org (%)		
		High	Medium	Low
1	Achieve customer satisfaction through early and continuous delivery of valuable software.	60	24	16
2	Changing requirements should be welcomed, even late in development for the customer's competitive advantage	40	34	26
3	Working software should be delivered frequently, from a couple of weeks to a couple of months, with a preference to the shorter timescale	25	42	33
4	Working software should be the primary measure of progress	37	34	29
5	Business people and developers should work together frequently throughout the project.	84	13	3
6	Projects should be built around motivated individuals who should be provided with the appropriate environment and support to get the job done	53	39	8
7	Face-to-face conversation is the most efficient and effective method of conveying information to and within a development team	60	26	14
8	The needs of the development team must be prioritized in line with cost, quality and time	35	18	47
9	The principles of good design should be strived for at all times	44	28	28
10	A development environment and its products should strive for and exhibit simplicity.	23	48	29
11	Architectures and designs should emerge and not be imposed.	26	21	53
12	Frequent reflection should be used for continuous improvement of development process and product.	23	44	33

The results also indicated that architecture and design were viewed from a top-down perspective and not as artifacts that should emerge from practice and dialogue. While this is interesting, the respondents' comments indicate a poor understanding of the

question and we have thus discounted data in this regard. Lastly, other interesting but less obvious results are:

- Project/process reflection has some organizational ‘presence’. Here practice results show that the primary techniques are ‘lessons learnt’ analysis (50%), self-reflection (24%) and as an outcome of formal monitoring and measuring (13%). Those respondents who stated that no explicit reflection was carried out cited time limitation as the cause.
- Documentation is still seen as important. Here, however, the majority of respondents viewed their current levels of documentation as necessary and sufficient for need (50%) though a significant number saw their documentation as necessary but not sufficient for their needs (38%). In correlation terms, the majority of respondents who note documentation as both necessary and sufficient use formal but tailored development processes.

Table 3: Agile themes in relation to their adoption in practice

Agile Themes	Relating Principles	Rating the take up of principles in Practice	Justification of Rating
Communication and Collaboration	7, 5	High	Both the frequency of communication and mode of communication being practiced adhere to Agile principles of daily face-to-face communication.
Team Involvement	6, 8	High	96% of the individuals are empowered to make decisions and over 64% take part in tailoring the process but team skills is the least used criteria/characteristic to tailor software development processes.
Reflection	12	Medium	89% of the respondents carry out some form of reflection, but only at the end of the project not throughout the project.
Frequent delivery of working software	1, 3, 4	Medium	The range of frequency in delivering working software to customers is wide (between 2 weeks – 3 months). In addition, 17% of the respondents only deliver on the final release. Project Milestones and Deliverables still remain the primary measure of progress (54%) while only 27% of the respondents measure progress through working software.
Managing Changing Requirements	2	Low	Although 99% of the respondents cater for change, only 38% do so using dynamic prioritization through iterative and incremental development.
Design	9, 10, 11	Void	These principles were not well understood by respondents.

Table 3 provides a summary of agile themes that are relevant in terms of their acceptance in practice. Overall, the findings suggest that there is an acceptance in the principles of agile development, but that this acceptance is implicitly tied with the acceptance of process tailoring and iterative lifecycles – the correlation between these factors and the agile principles seen as important in practice is strong. This provides support for other studies noting that, to

be agile, a project’s processes must be cyclical and involve builds and increments done in parallel [8].

6. Conclusion

All surveys carry health warnings and this example is not immune. Overall, the sample size of the data can be considered limited in that it addresses a UK audience and the response rate is small at 6%. With such limitations in mind the results have been presented in order to allow the readers to draw their own implications. Overall, however, a number of interesting conclusions may be drawn in relation to the state of practice and the implicit use of agile principles:

- A number of respondents state that they develop software in an ad-hoc manner (26%). From one perspective, this finding may be considered worrying in light of the ongoing issues of the software crisis. From another perspective, however, ad hoc does not necessarily equate to disorganized and may be related to the use of ‘rules of thumb’ used implicitly across development (see [21]), where formal process is considered too heavy. Indeed, a significant number of respondents in this respect work in small development environments (63% have fewer than 9 employees).
- The trend of adapting/tailoring software development process to context is on the increase. In relation to an earlier study (see [22]), the findings here suggest an increase in the number of organizations that tailor/adapt their development process and a decrease in the number of organizations that follow a commercial process in a prescriptive fashion.
- A large number of organizations now base their software development on iterative and/or incremental lifecycles (68%). The frequency of software delivery ranges from every two weeks to every three months (the majority) and by far outweighs delivery only on final release. Change control, however, is still firmly locked in a traditional project management loop

In summary, this research aimed to: (a) to review current software development practice; (b) to assess perceptions related to agile principles and values and (c) to examine the implicit/explicit integration of agile principles with mainstream development/organizational practice.

In reviewing current software development practice, it can be concluded that the adoption of software development methods is increasing as opposed to findings in [22] which highlighted the decrease of software development usage in practice. However, the type of software development methods being adopted have changed from traditional, prescriptive software development methods that follow the waterfall lifecycle to software development frameworks that require tailoring to specific contexts and that follow iterative and incremental lifecycles (e.g. RUP and DSDM).

In relation to the second objective of this research, one can conclude that a majority of practitioners perceive agile values and principles as important. The Agile Manifesto has just made these values and principles explicit and now a more conscience effort can be put towards implementing such values and principles in practice (e.g., time for reflection can be accounted for, frequency of software delivery and managing change can become a priority).

When examining the implicit/explicit integration of agile principles with mainstream development/organizational practice, the principles that were rated of high importance were implemented in practice as well. Those principles influenced by traditional prescriptive organizational processes (e.g. managing frequent delivery of software through dynamic prioritization as opposed to change request processes), on the other hand, were not being implemented in practice.

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